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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

4084400163

MATHEMATICS (SYLLABUS D) **4024/21**

Paper 2 **October/November 2011**

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments
Electronic calculator

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Section A
Answer **all** questions.

Section B
Answer any **four** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.
You are expected to use an electronic calculator to evaluate explicit numerical expressions.
If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.
For π, use either your calculator value or 3.142, unless the question requires the answer in terms of π.

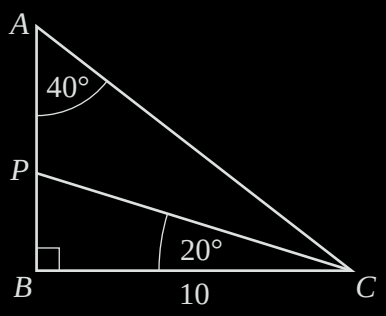
The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 100.

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Calculate

(a) *PB*,

Answer ..... cm [2]

(b) *AP*,

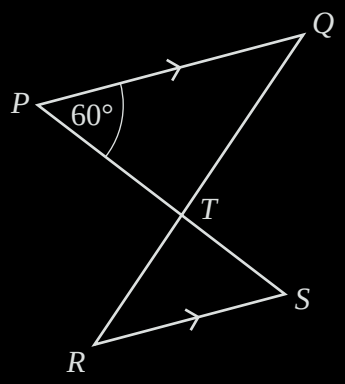
Answer ..... cm [2]

(c) the perimeter of triangle *PBC*.

Answer ..... cm [3]

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(a) Write down the value of $P\hat{S}R$. Give a reason for your answer.

Answer $P\hat{S}R$ = ... because .....

..... [1]

(b) The bearing of Q from P is 070°.
Find the bearing of

(i) S from P,

Answer ..... [1]

(ii) P from S,

Answer ..... [1]

(iii) R from S.

Answer ..... [1]

(c) (i) Explain why triangles PQT and SRT are similar.

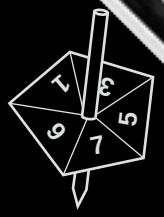
Answer .....

..... [1]

(ii) Given that PT = 54 m, TS = 36 m and RQ = 85 m, find TQ.

Answer ..... m [3]

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+	1	3	5	7	9
1	2	4			
3					12
5					
7				14	
9					

(i) Complete the possibility diagram. [2]

(ii) Find the probability that the sum of the two numbers is

(a) odd,

Answer ..... [1]

(b) 14 or more.

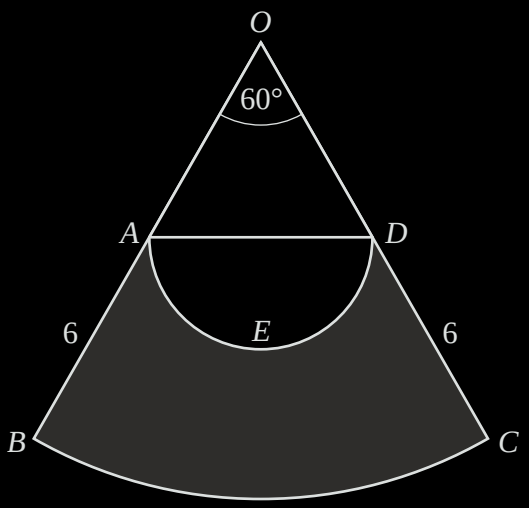
Answer ..... [1]

(c) Katrina spins it three times.

Calculate the probability that the three numbers obtained are the same.
Express your answer as a fraction in its lowest terms.

Answer ..... [2]

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In the diagram, OBC is the sector of a circle, centre O, and $B\hat{O}C = 60°$.
A and D are the midpoints of OB and OC respectively, and $AB = DC = 6$ cm.
AED is a semicircle with AD as diameter.

(a) Show that $AD = 6$ cm.

[1]

(b) The length of the arc BC is $n\pi$ centimetres.

(i) Find n.

Answer ..... [1]

(ii) Find $\dfrac{\text{the length of the arc } AED}{\text{the length of the arc } BC}$.

Answer ..... [2]

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LAWNMOWER

$2395



Plan A: Deposit $595 and
12 monthly payments of $171.04

Plan B: Deposit $395 and
24 monthly payments of $

(i) Rashid buys one of these lawnmowers for $2395.
Sayeed buys one of these lawnmowers using Plan A.

In total, how much **more** than Rashid will Sayeed pay?

Answer $..... [1]

(ii) When one of these lawnmowers is bought using Plan B, the total cost is $3054.20.

Calculate the monthly payment.

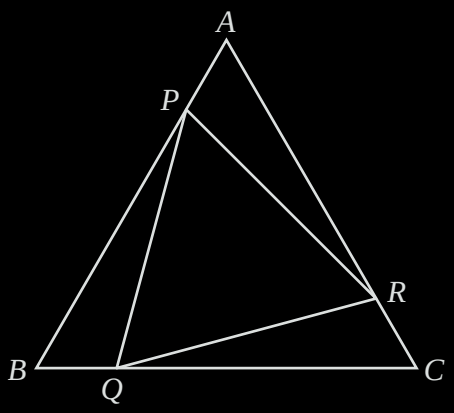
Answer $..... [2]

(iii) In a sale, the price of the lawnmower is reduced from $2395 to $1595.

Calculate the percentage discount.

Answer ..... % [2]

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In the diagram, ABC is an equilateral triangle.
The points P, Q and R lie on AB, BC and CA respectively, such that $AP = BQ = CR$.

(i) Show that triangles APR, BQP and CRQ are congruent.

[3]

(ii) It is given that $AB = 5\,\text{cm}$ and $PQ = 4\,\text{cm}$.

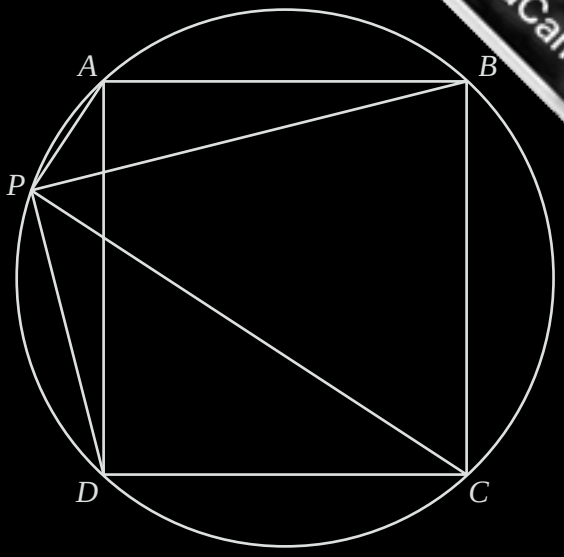
(a) Find $\dfrac{\text{Area of triangle } PQR}{\text{Area of triangle } ABC}$.

Answer .....[1]

(b) Find $\dfrac{\text{Area of triangle } APR}{\text{Area of triangle } ABC}$.

Answer .....[1]

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(i) Explain why $A\hat{P}C = 90°$.

Answer ..... [1]

(ii) Explain why $A\hat{P}B = B\hat{P}C$.

Answer ..... [2]

(iii) Hence find

(a) $A\hat{P}B$,

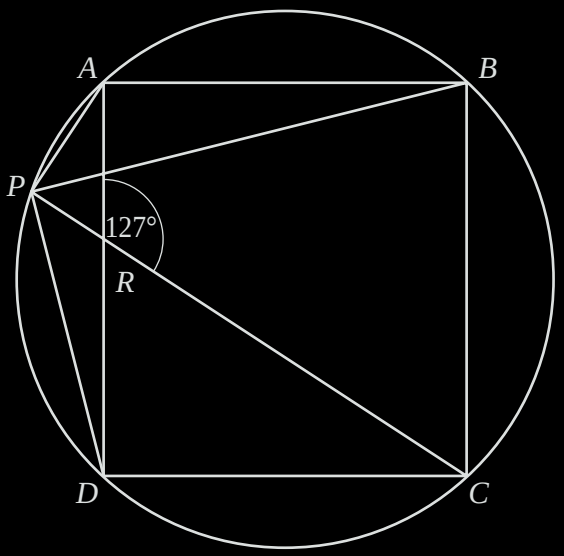
Answer ..... [1]

(b) $A\hat{P}D$.

Answer ..... [1]

(iv) PC and AD intersect at R.

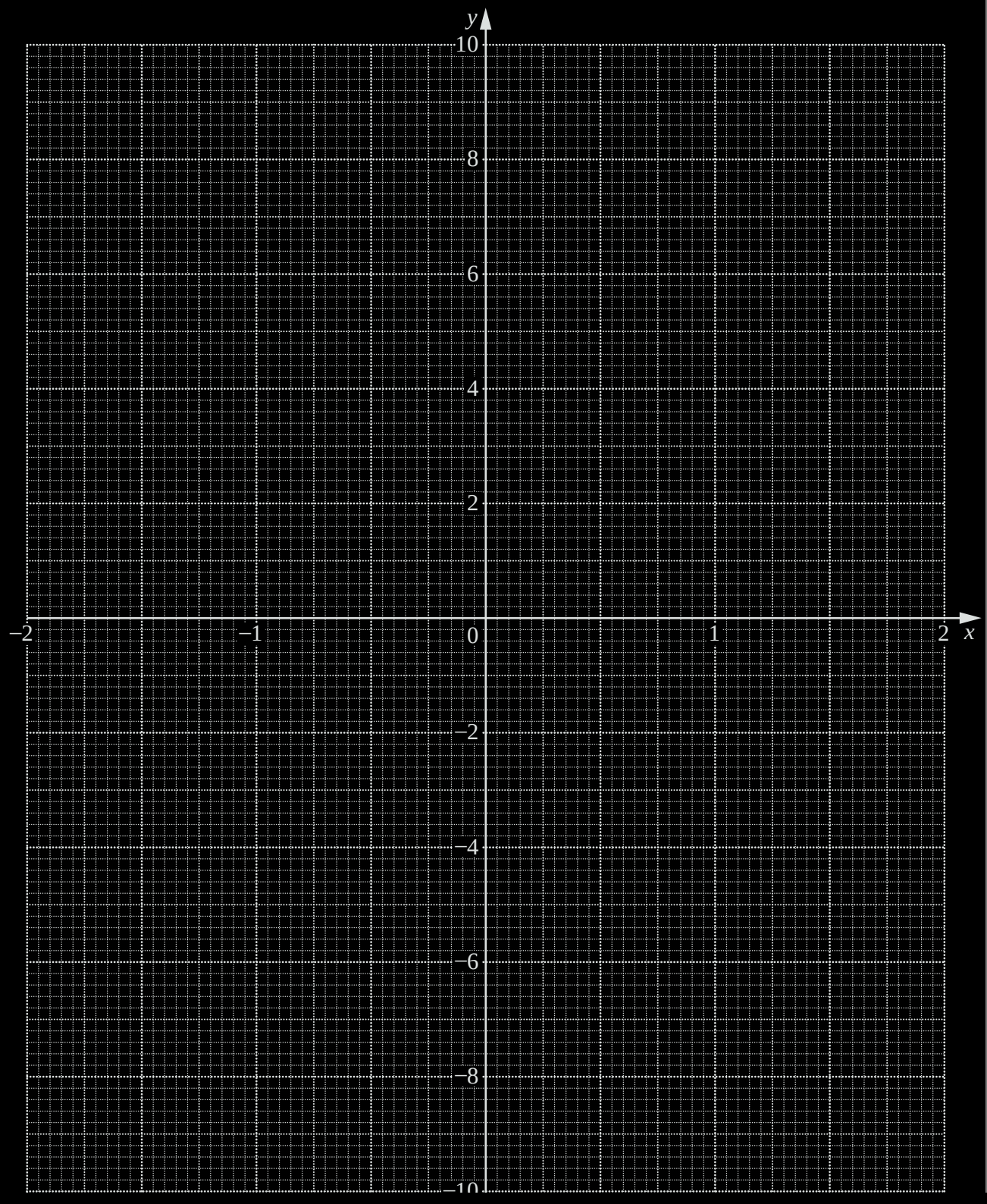
Given that $A\hat{R}C = 127°$, find $P\hat{D}C$.



Answer ..... [2]

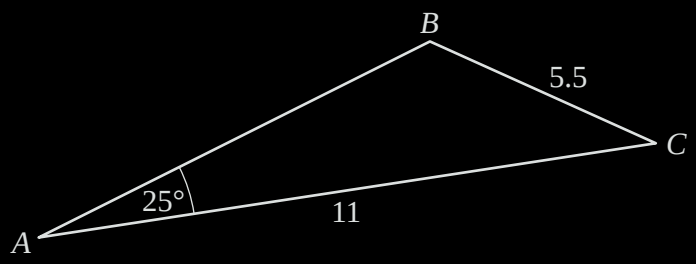
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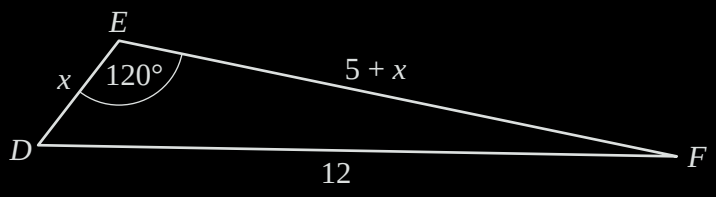


In the diagram, $AC = 11$ cm, $BC = 5.5$ cm and $B\hat{A}C = 25°$.
It is given that $A\hat{B}C$ is an **obtuse** angle.

Calculate $A\hat{B}C$.

Answer ..... [4]

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In the diagram, $DF = 12$ cm, $DE = x$ centimetres and $EF = (5 + x)$ centimetres.

(i) Form an equation in x and show that it reduces to $3x^2 + 15x - 119 = 0$.

[3]

(ii) Solve the equation $3x^2 + 15x - 119 = 0$, giving each answer correct to 3 decimal places.

Answer $x =$ ..... or ..... [4]

(iii) Find the length of EF in millimetres, correct to the nearest millimetre.

Answer ..... mm [1]

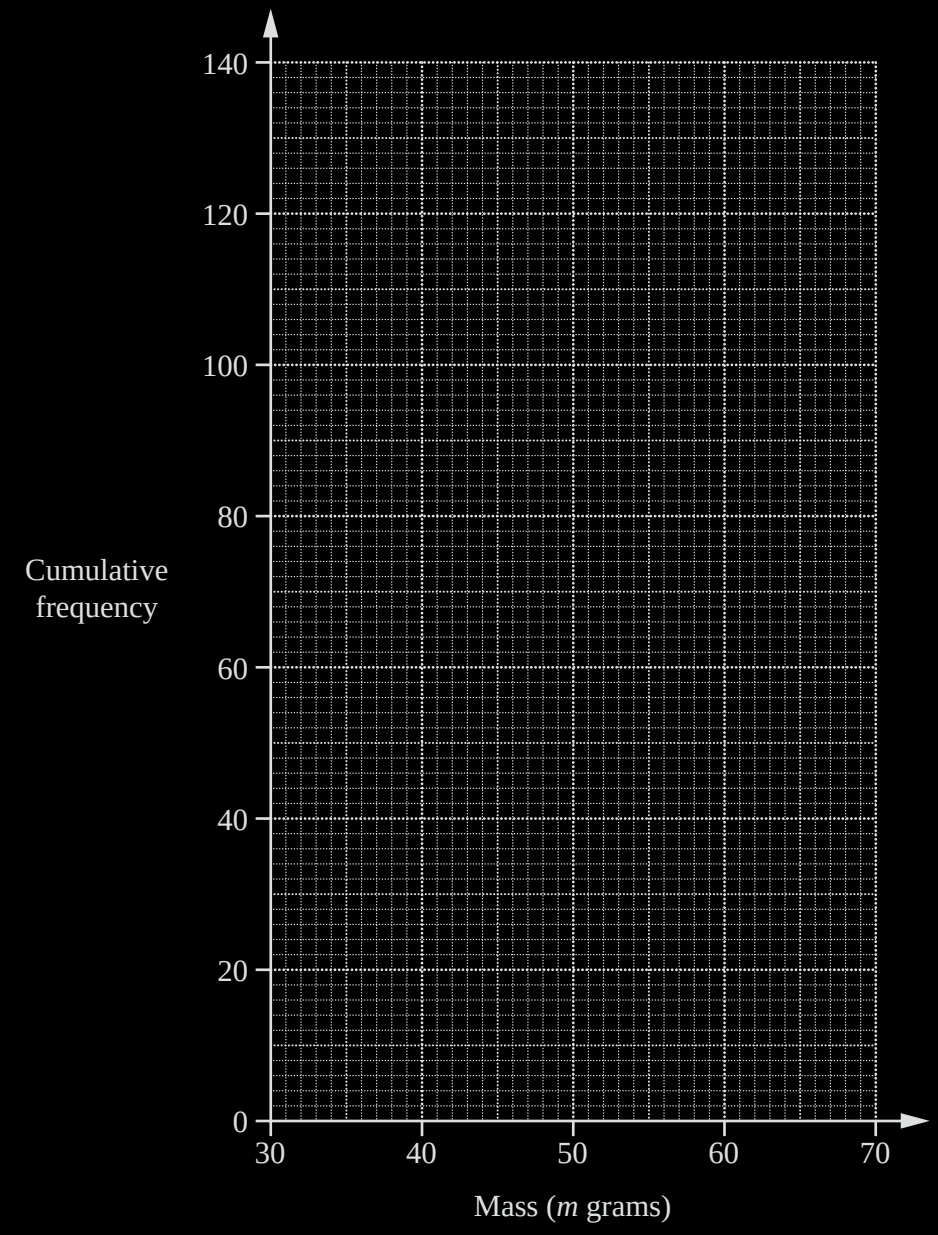
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	≤	≤	≤	≤	≤	≤

≤ ≤

	≤	≤	≤	≤	≤	≤	≤

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[3]

(c) Use your graph to find

(i) the median mass of the eggs,

Answer ..... g [1]

(ii) the interquartile range.

Answer ..... g [2]

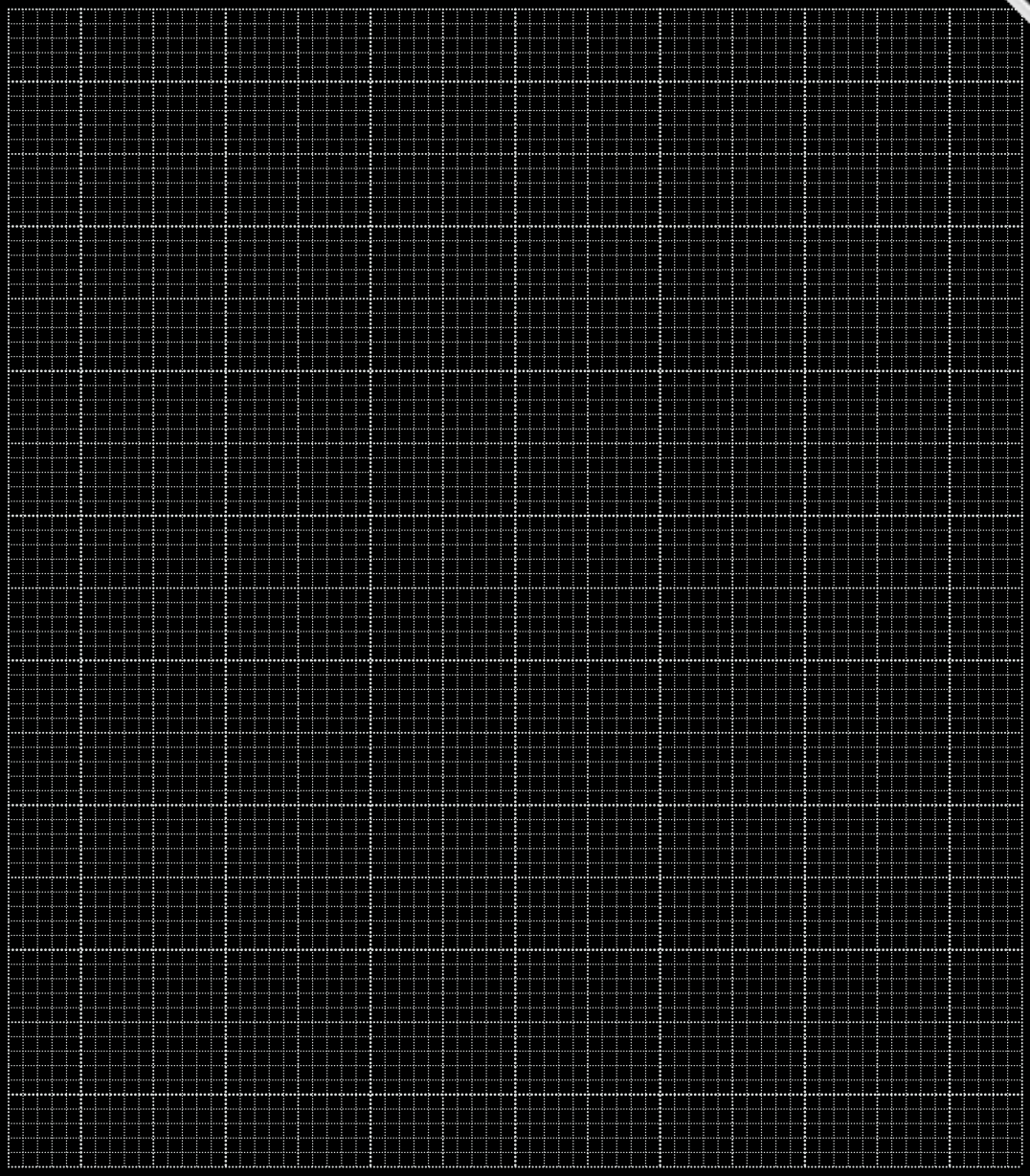
(d) The 12 eggs with the greatest mass are classed as extra large.
The 30 eggs with the least mass are classed as small.

Use your graph to find an estimate of the smallest difference in mass between an extra large egg and a small egg.

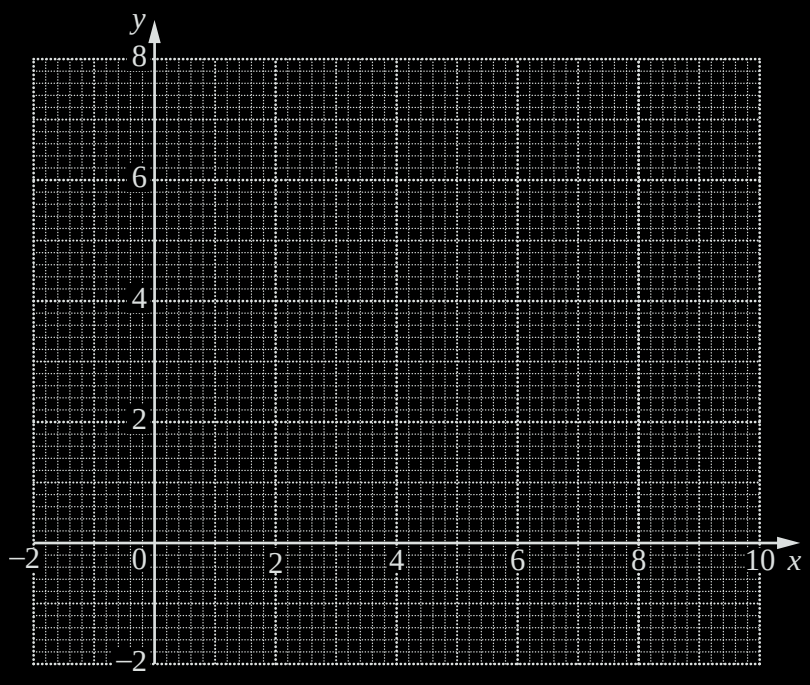
Answer ..... g [2]

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